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Key Points:

- Zircon and whole-rock crustal thickness proxies align with field evidence for extensional and contractional deformation
- Modest Late Cretaceous crustal thinning coincided with normal faulting and slower, more oblique plate convergence
- Crust remained thin during Abanico deposition (~37–17 Ma); ~10 km thickening occurred during Neogene shortening

Supporting Information:

Supporting Information may be found in the online version of this article.

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Building the Andean Crust at ~35°S: Arc Geochemistry Reveals Cretaceous Thinning and Rapid Neogene Thickening

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Abstract Quantification of the timing and mechanisms of crustal thickening in Cordilleran orogenic hinterlands remains challenging. We address this in the Andes using whole-rock geochemistry and zircon petrochronology from a >70 Myr volcanic record in the Tinguiririca valley of Chile (~35°S), which captures a pivotal transition from extensional to contractional deformation. Zircon (Sm, Gd, Dy)/Yb ratios generally track trends in whole-rock geochemical crustal thickness (paleomohometry) models across this record, with Dy/Yb showing the most consistent correspondence. Our results indicate modest crustal thinning coeval with Late Cretaceous normal faulting and oblique plate convergence. Eocene and Early Miocene samples yield similar crustal thicknesses (~35 km), suggesting limited local net attenuation during proposed intra-arc rifting (Abanico basin). The crust then thickened by ~10 km to near-modern values (~46 km) during the Neogene; Quaternary estimates match modern geophysical constraints, validating the paleomohometry approach. Thickening was most rapid between ca. 17 and 13 Ma, coinciding with accelerated retroarc shortening and plate convergence. Additional zircon geochronologic and geochemical proxies (Th/U, phosphorus content, and oxygen fugacity) indicate enhanced Miocene crustal melting and magmatic productivity. Along-strike comparisons reveal north-to-south younging and decreasing magnitude of thickening, mirroring trends in retroarc shortening. These findings indicate that underthrusting of foreland lithosphere during retroarc shortening drove crustal thickening, magmatic flare-up, and generation of orogenic topography in the southern Central Andes.

Plain Language Summary The Andes are one of Earth's great mountain belts, built as tectonic forces thickened the crust beneath them. Yet geologists have struggled to pinpoint when and how quickly this thickening occurred. We used the chemistry of volcanic rocks spanning over 70 million years in central Chile to track changes in crustal thickness. Our results reveal that the crust remained relatively thin in this area until less than 20 million years ago, after which faster convergence of an oceanic plate pushed South American continental material beneath the growing Andes. We found that this process likely caused increased volcanic activity and rapid thickening of the crust below the mountain belt. By comparing chemical estimates with independent evidence from field geology, we validated which geochemical tools reliably record crustal thickness changes, which will aid in studies of ancient mountain belts worldwide.

1. Introduction

Continental magmatic arcs are key sites of crustal growth and important archives of Cordilleran-type orogenic processes, preserving geochemical records of evolving melt source and crustal thickness through time (e.g., Ducea et al., 2015; Kay et al., 2005; Luffi & Ducea, 2022b; Profeta et al., 2015). Long-lived magmatic archives are rare in shallowly exhumed orogens, however, where the ancient record is often obscured by younger volcanism and deformation. Consequently, most previous investigations of long-term Andean arc evolution have averaged data across large areas (>15,000 km²), making it difficult to link geochemical trends to local geologic constraints on structural development (e.g., Bowman, 2024; Kay et al., 2005; Muñoz-Gómez et al., 2025; Profeta et al., 2015). The Tinguiririca valley of Chile (~35°S) preserves an exceptionally well-exposed, volcanic-dominated stratigraphic sequence spanning the Late Cretaceous to Quaternary, providing an opportunity to

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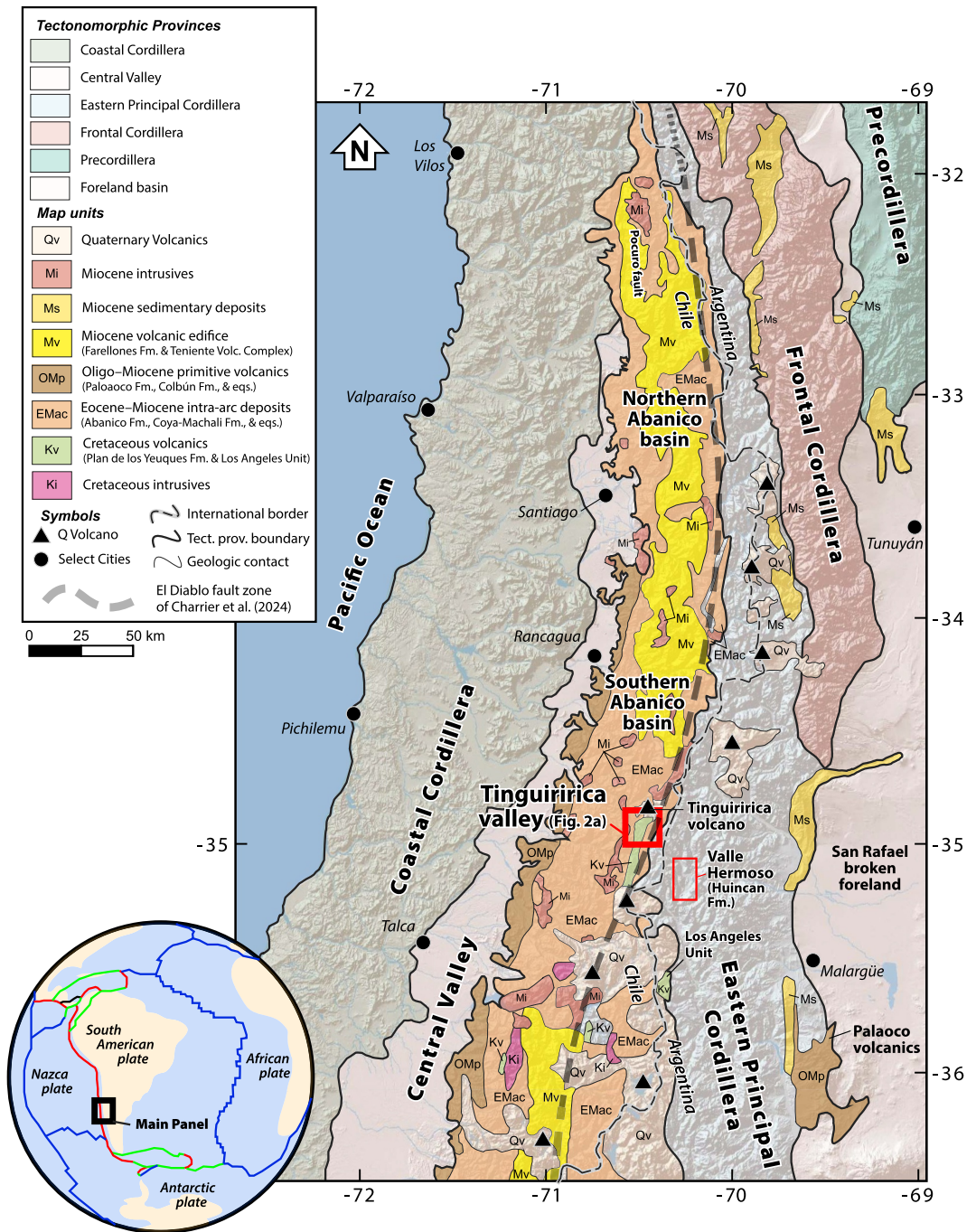


Figure 1. Tectonomorphic map of the Abanico basin and surrounding region, showing distribution of Cretaceous–Miocene rock units discussed in the text. Other rock units are grouped by tectonomorphic province. Unit divisions modified from Piquer et al. (2017) and Charrier et al. (2024). Contacts based primarily on Charrier et al. (2024) and references therein, with additions and modifications from Piquer et al. (2017), Fennell et al. (2019), and new mapping. Inset globe shows tectonic context of Andes, with blue, green, and red lines respectively representing divergent, transform, and convergent plate boundaries. Legend abbreviations: eqs.—equivalent units; tect. prov. boundary—tectonomorphic province boundary.

better investigate these relationships (Figures 1 and 2; e.g., Mosolf et al., 2018; Muñoz et al., 2018). This volcanic record captures a shift from extension to contraction and the inception of major Andean mountain building at this latitude (e.g., Charrier et al., 2015, 2024; Fennell et al., 2019; Horton et al., 2016; Iannelli et al., 2018). Additionally, it occupies a key position between the high Central Andean Plateau with thick crust (≥ 70 km) to the north

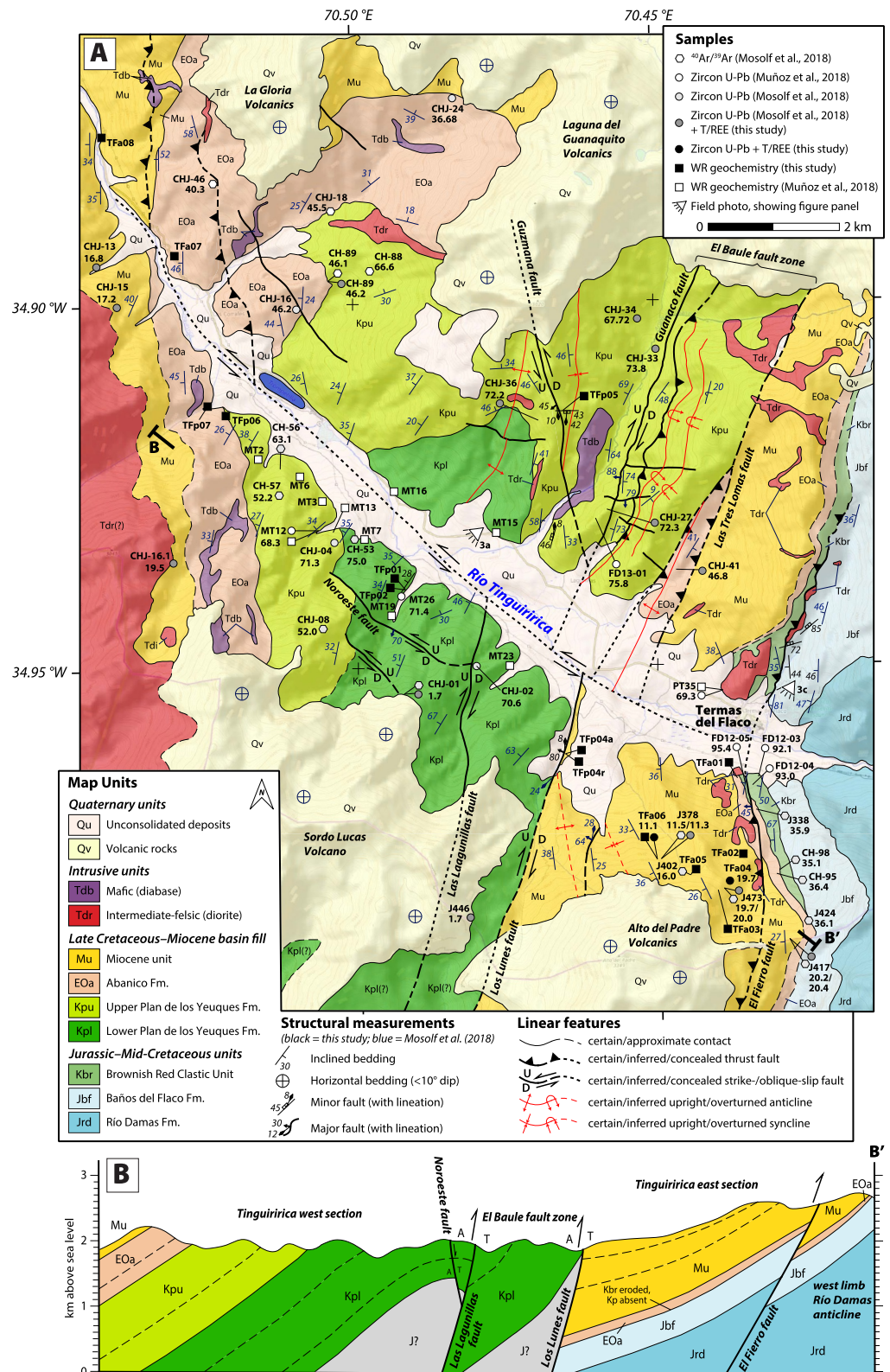


Figure 2. (a) Geologic map of the upper Tinguiririca valley area, modified from Mosolf et al. (2018) to incorporate observations and samples of this study and Muñoz et al. (2018). Samples are labeled with sample name and age determined from the indicated geochronometer. (b) Geologic cross-section along the southern side of the Tinguiririca valley, modified from Mosolf et al. (2018). Surficial and intrusive units are not shown. Scale is same as map, with no vertical exaggeration.

and the lower Patagonian Andes with thinner crust (≤ 40 km) to the south (e.g., Isacks, 1988; Kley et al., 1999; Tassara & Echaurren, 2012). These attributes make the Tinguiririca area an important link in resolving the spatiotemporal evolution of Andean magmatism and sub-arc crustal thickness.

Outstanding questions remain regarding the history of crustal thickness along the southern Central Andes: did the crust significantly thin during proposed Late Cretaceous–Paleogene extension (e.g., Charrier et al., 2024; Fennell et al., 2019; Fernández Paz et al., 2024; Muñoz et al., 2018; Muñoz-Gómez et al., 2025; Nyström et al., 2003) or remain relatively stable as implied by transpressional models (e.g., Cobbold & Rossello, 2003; Mosolf et al., 2018)? When and why did the crust thicken, and what was the relative contribution from horizontal shortening versus other mechanisms (i.e., magmatic addition, crustal flow, underplating; e.g., Anderson et al., 2017; Carrapa et al., 2022; Kay et al., 2005; Kley & Monaldi, 1998; Isacks, 1988; Ouimet & Cook, 2010)? When did the along-strike crustal thickness gradient arise (>25 km decrease from 20 to 35°S), and how does it relate to variations in fold-thrust belt development (e.g., Giambiagi et al., 2022; Haschke et al., 2002; Horton, 2018; Kley et al., 1999; Isacks, 1988; Tassara & Echaurren, 2012)?

We address these questions through a geochemical investigation of the Cretaceous–Quaternary volcanic succession preserved in the Tinguiririca area, evaluated alongside independent structural and geophysical constraints. We apply paleomohometry models, using empirical calibrations relating arc magma geochemistry to sub-arc Moho depth (Luffi & Ducea, 2022b), to whole-rock igneous samples (13 new, 22 compiled). We also test proposed petrochronologic proxies for crustal thickness and magmatic evolution using comagmatic zircon from volcanoclastic samples, integrating U–Pb age and trace-element measurements (294 analyses from 22 samples; e.g., Balica et al., 2020; Tang et al., 2021). We then evaluate these data against crustal thickness reconstructions from area-balanced cross sections at the same latitude (Giambiagi et al., 2022).

Our multi-proxy approach reveals critical changes in crustal thickness that provide new constraints on models of Cretaceous–Paleogene extension and support the dominant role of Neogene shortening in building the southern Central Andes. We find agreement between geochemical and structural constraints (Giambiagi et al., 2022), enabling examination of how shortening, magmatic productivity, and crustal thickness covary during orogenesis. Along-strike comparisons reveal variations in timing and magnitude of crustal thickening between the Central Andean Plateau and southern Central Andes, suggesting that thickening decreases in both magnitude and age from north to south (Bowman, 2024; Muñoz-Gómez et al., 2025). These findings provide insight into feedbacks between tectonics and magmatism, the viability of geochemical crustal thickness proxies, and the processes generating orogenic topography in Cordilleran systems.

2. Regional Tectonic Context

2.1. Tectonomorphic Framework

The Andes are a $>7,000$ km-long mountain belt formed above an east-dipping subduction zone along the western margin of South America. At the latitude of the study area ($\sim 35^\circ$ S), the Nazca oceanic slab presently dips $\sim 27^\circ$ and converges at ~ 7 cm/yr (e.g., Hayes et al., 2018; Kendrick et al., 2003). This subduction generates arc magmatism of the southern Andean volcanic zone, locally represented by the Quaternary Tinguiririca volcano (Figure 1). Geophysical data indicate modern crustal thickness of ~ 44 – 46.5 km in this segment (Laske et al., 2013; Tassara & Echaurren, 2012).

At this latitude, the orogen consists of five tectonomorphic provinces, from west to east: the Coastal Cordillera, consisting of a panel of Paleozoic–Mesozoic metamorphic and igneous rocks in western Chile; the Chilean Central Valley forearc basin; the Western Principal Cordillera (WPC), where Late Cretaceous–Miocene volcanic rocks are deformed in the western part of the fold-thrust belt; the Eastern Principal Cordillera (EPC), consisting of Permian–Cenozoic rocks deformed in the Malargüe fold-thrust belt; and an eastward adjacent foreland basin system, segmented by back-arc volcanic centers and the San Rafael block uplift (Figure 1; e.g., Farías et al., 2010; Giambiagi et al., 2012; Hervé et al., 2013; Horton et al., 2016; Mpodozis & Ramos, 1989). The Mesozoic–Cenozoic succession of the Tinguiririca valley crops out for ~ 10 km eastward from the WPC–EPC boundary (Figures 1 and 2).

2.2. History of Orogenic Development

The Andean orogen at the latitude of this study records alternating upper plate extension and contraction in response to Mesozoic–Cenozoic subduction (e.g., Charrier et al., 2015; Fennell et al., 2019; Mpodozis & Ramos, 1989). Arc magmatism has been active since at least Triassic time, initially accompanied by back-arc extension (e.g., Charrier et al., 2015). An early contractional phase has been proposed during the Cenomanian–Santonian (ca. 100–85 Ma) based on synorogenic sedimentation in the study area and eastward adjacent regions of Argentina (e.g., Balgord & Carrapa, 2016; Di Giulio et al., 2012; Fennell et al., 2017, 2020; Mescua et al., 2013; Muñoz et al., 2018; Tunik et al., 2010). However, the magnitude of orogenic growth during this interval remains debated, with some models favoring only moderate, localized shortening (Encinas et al., 2024). Subsequent Campanian magmatism occurred coevally with localized normal faulting in the study area (Muñoz et al., 2018) and deposition of the Colimapu Formation at 33–34°S under hypothesized extensional conditions (Henríquez et al., 2024). This extension has been linked to slab roll-back and/or opening of an asthenospheric window during subduction of the spreading ridge separating the ancestral Farallon and Aluk (Phoenix) oceanic plates (e.g., Iannelli et al., 2018; Muñoz et al., 2018).

During Late Eocene–Early Miocene time, the Abanico basin and correlative intra-arc depocenters developed across ~31–44°S (e.g., Charrier et al., 2024; Horton et al., 2016). Geochemical trends and reported normal fault-related growth strata have generally been interpreted to reflect extensional conditions (e.g., Charrier et al., 1996, 2002, 2005, 2007; Fernández Paz et al., 2024; Muñoz-Gómez et al., 2025; Muñoz-Sáez et al., 2014; Muñoz et al., 2006; Nyström et al., 2003; Piquer et al., 2017) and potential opening of a major intra-arc graben between ca. 36 and 18 Ma (Charrier et al., 2024). The proposed extensional regime contrasts with localized evidence of coeval transpressional deformation in the region (Cobbold & Rossello, 2003; Mosolf et al., 2018) and sustained shortening documented north of 24°S over the same interval (e.g., DeCelles et al., 2011; Henriquez et al., 2022).

Retroarc shortening is widely recognized by Early Miocene time. Thermochronologic data indicate hinterland exhumation at ca. 20 Ma, and foreland basin deposits record rapid subsidence and arrival of thrust-belt-derived detritus after ca. 19 Ma (e.g., Bande et al., 2020; Horton et al., 2016; Howlett, 2025). The thrust front propagated eastward to near its present position between ca. 10 and 6 Ma, recorded by growth strata deposited atop leading thrusts (e.g., Fuentes et al., 2016; Horton et al., 2016). Balanced cross sections and kinematic models show 28–46 km of crustal shortening across the retroarc fold-thrust belt, accommodated mainly during the Middle–Late Miocene (e.g., Giambiagi et al., 2022; Howlett, 2025).

3. Geology of the Tinguiririca Area

3.1. Stratigraphy

The chronostratigraphy of the Tinguiririca valley section is constrained by 39 dated lava and tuff samples. Figure 3 shows composite stratigraphic sections synthesizing lithologic observations and age control from Mosolf et al. (2018), Muñoz et al. (2018), and this study (see Section 4.2). Details on this synthesis and compiled sample information are reported in Text S1 in Supporting Information S1 and Data Set S1, respectively.

The stratigraphically lowest exposed deposits consist of >4 km of volcanic and continental sedimentary rocks of the Upper Jurassic Río Damas Formation and ~400 m of mainly marine rocks of the Lower Cretaceous Baños del Flaco Formation, recording deposition in an extensional back-arc basin (Figures 1 and 2; e.g., Charrier et al., 2015; Junkin & Gans, 2019; Kohn, 1960). The older Mesozoic units are locally overlain by ~300 m of fluvial deposits of the Brownish Red Clastic Unit (BRCU), interpreted to record foreland basin deposition during Cenomanian–Santonian shortening (e.g., Charrier et al., 1996; Mescua et al., 2014; Muñoz et al., 2018). Above the BRCU lies an up to ~4 km thick package of Upper Cretaceous to Miocene, mainly volcanic and volcanoclastic strata, studied here. We divide this package into the Campanian–Danian(?) Plan de los Yeuques Formation, Eocene–Oligocene(?) Abanico Formation, and overlying rocks referred to as the Miocene unit, following the stratigraphic framework of Muñoz et al. (2018).

The Plan de los Yeuques Formation is present only west of the El Baule fault zone—a major south-southwest-striking structural boundary that controls stratigraphic variations across the study area (Figure 2). The formation consists primarily of andesitic and basaltic lava intercalated with volcanic breccia (ca. 80–52 Ma; ~1.6 km exposed thickness; Figure 3). The overlying Abanico Formation consists mainly of lahar deposits, tuffs, and andesitic to basaltic lavas deposited between ca. 52 and 20 Ma regionally, with local accumulations constrained

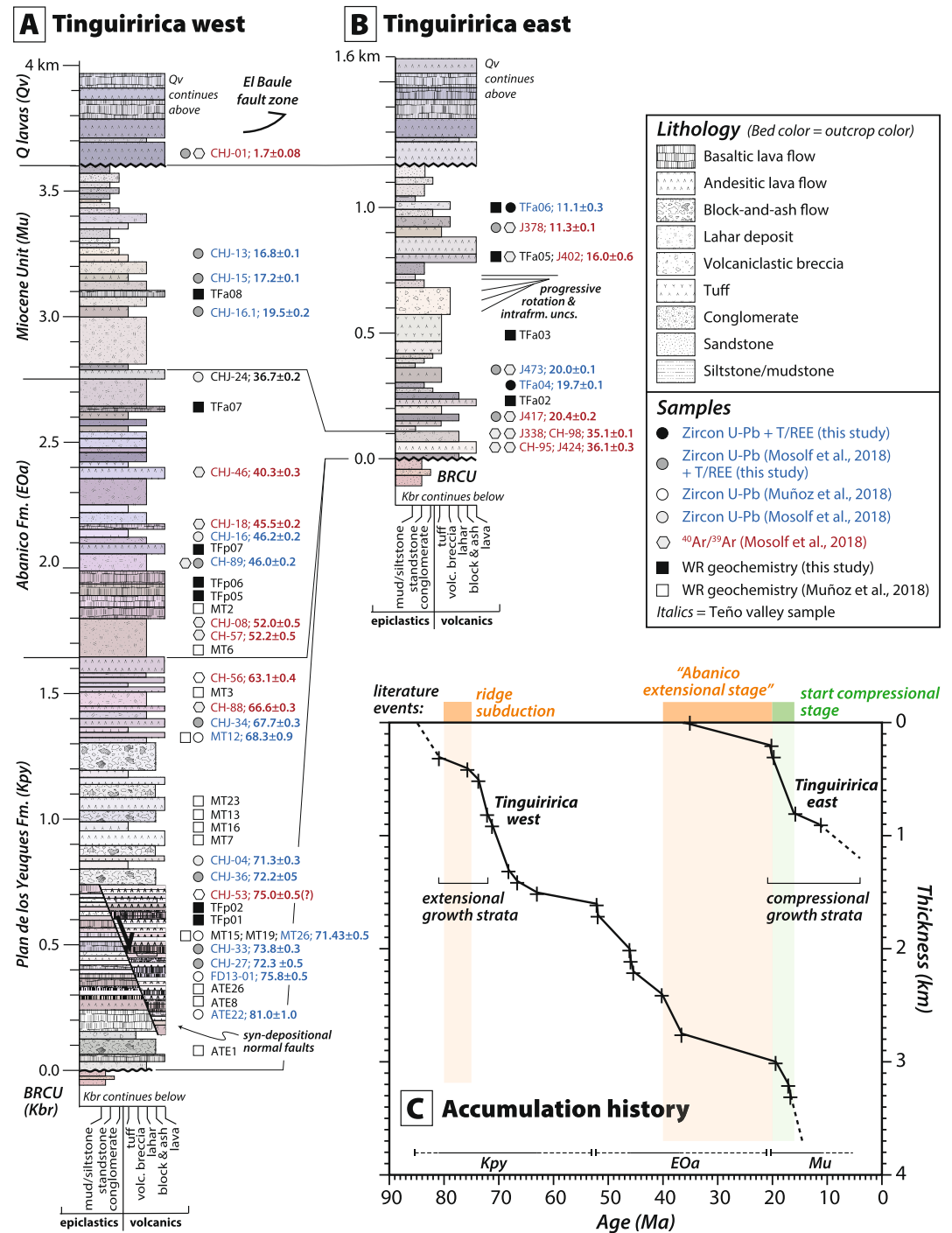


Figure 3. Composite stratigraphic section of Late Cretaceous–Quaternary rocks in the Tinguiririca area, (a) west and (b) east of the El Baule fault zone. Sections show lithology using conventional fill patterns (legend at upper right; fill color corresponds to outcrop color). The stratigraphic positions of samples were synthesized from published sections, map patterns, and field observations. Eruption and maximum depositional ages were recalculated from zircon U-Pb data by the maximum likelihood age (MLA) method (Vermeech, 2021). $^{40}\text{Ar}/^{39}\text{Ar}$ ages are from Mosolf et al. (2018). The lower portion of the western section was projected from the adjacent Teno valley, based on Muñoz et al. (2018, samples shown in italics). Intervals previously interpreted to contain extensional and compressional growth strata are indicated (intrafm. uncs.—intraformational angular unconformities; after Charrier et al., 1996; Mosolf et al., 2018; Muñoz et al., 2018). (c) Cumulative stratigraphic thickness through time for Tinguiririca west and east sections, uncorrected for compaction. Points represent select geochronologic control from (a) and (b). Tectonic events from the literature are annotated (Charrier et al., 2024; Horton et al., 2016, and references therein).

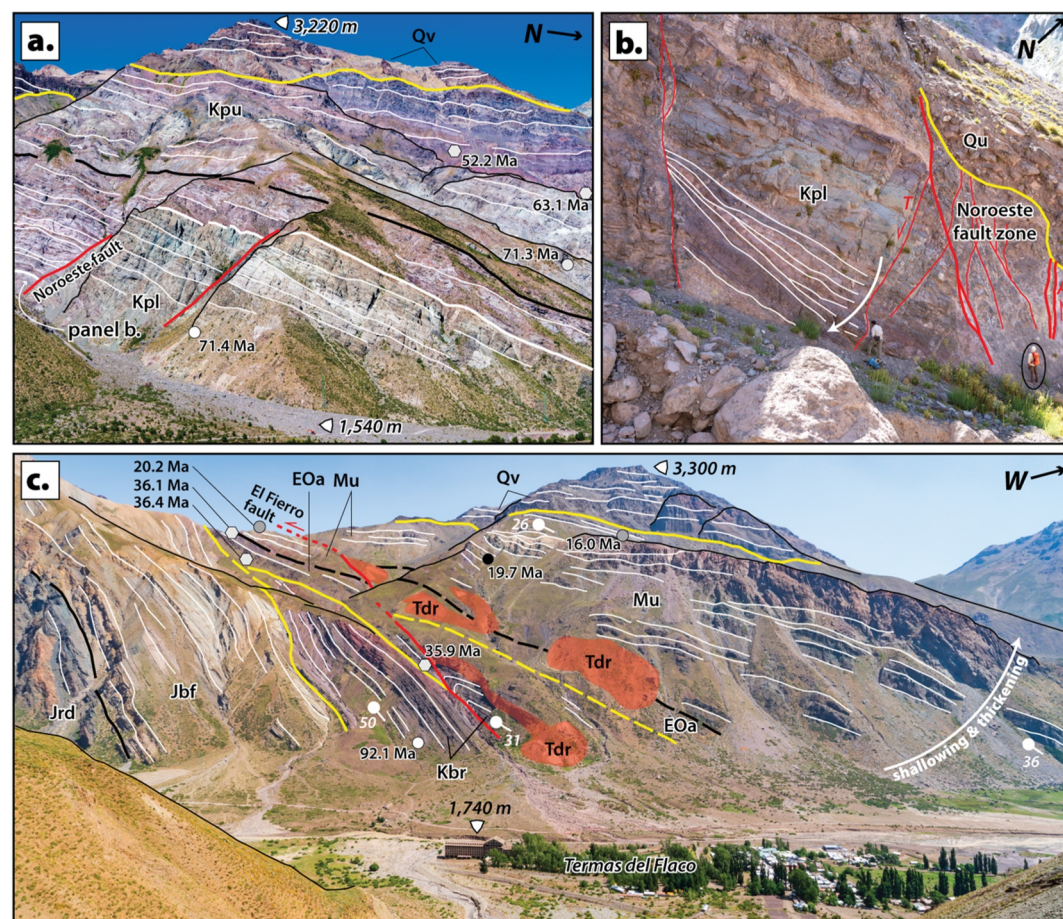


Figure 4. Field photos showing evidence of syndepositional deformation in the study area. (a) The Noroeste fault accommodates a greater thickness of the lower Plan de los Yeuques Formation in its hanging wall (Muñoz et al., 2018) and is sealed by overlying ≤ 72 Ma volcanic deposits. (b) Volcaniclastic sandstones adjacent to this fault zone progressively thicken northeastward toward the master splay. Geologist circled for scale. (c) The Abanico Formation overlies the Cretaceous Brownish Red Clastic Unit along a gentle angular unconformity east of the El Baule fault zone, and both formations are obliquely cut by the El Fierro fault. Beds of the overlying Miocene unit progressively shallow up-section (white ball-and-ticks show measured dips) and thicken toward the basin center, interpreted to reflect syndepositional growth of the Río Damas anticline (Charrier et al., 1996; Mosolf et al., 2018). Yellow lines show unconformities; white lines highlight bedding; red lines show faults; thin black lines highlight ridges; and thick black lines show formation boundaries. Lines are dashed where covered/inferred. Select samples are shown. Sample symbology, unit abbreviations, and photo locations are shown in Figure 2.

largely to ca. 47–35 Ma (Figure 3). The Abanico Formation is ~ 1.1 km thick west of the El Baule fault zone, but only thin accumulations of the upper part (~ 100 m; ≤ 36.1 Ma) are present farther east, where they rest on an angular unconformity cutting through the BRCU and into the Baños del Flaco Formation (Figures 2 and 4c; Mosolf et al., 2018). The ca. 20 Ma to ≤ 11 Ma Miocene unit overlies the Abanico Formation throughout the study area and consists mainly of andesite, tuff, and lahar deposits, with sedimentary facies in its upper part (Figures 2 and 3; Mosolf et al., 2018). The older volcanogenic formations are unconformably overlain by Quaternary (≤ 1.7 Ma) lavas and tuffs throughout the Tinguiririca area.

3.2. Structure and Syndepositional Deformation

The most significant structures in the area are the south-southwest-striking, steeply-dipping faults of the El Baule fault zone, which exhibit dextral-reverse kinematics and $>1,200$ m of stratigraphic separation between Mesozoic–Cenozoic strata (Figure 2; Mosolf et al., 2018). These structures show mutual cross-cutting relationships with subordinate east-to-south-striking faults and are associated with upright, shallowly north-northeast plunging folds (Figure 2; Mosolf et al., 2018). Beds in the lower part of the Plan de los Yeuques

Formation exhibit >300 m of offset across associated steeply southwest-dipping faults, with greater stratal thickness in the hanging-wall (Figures 2, 4a, and 4b). These relationships were interpreted by Muñoz et al. (2018) as normal fault-related growth strata, indicating syndepositional extension. Encinas et al. (2024) similarly reported synextensional growth strata in the Plan de los Yeuques Formation at ~36°S.

In the eastern study area, the El Fierro fault cuts the BRCU, Abanico Formation, and the Miocene unit with <200 m of stratigraphic separation (Figures 2 and 4c; e.g., Charrier et al., 1996; Davidson, 1971; Giambiagi et al., 2019; Mosolf et al., 2018). Beds of the Miocene unit thicken westward and progressively rotate to shallower dips up-section (Figures 2b and 4c; Charrier et al., 1996). Mosolf et al. (2018) interpreted this relationship to record deposition atop the growing western limb of the Río Damas anticline—a km-scale fold with its axis near the international border with Argentina.

Tectonic interpretations of the El Baule and El Fierro faults differ in both timing and kinematics. Muñoz et al. (2018) emphasized evidence for Late Cretaceous extensional tectonics, including normal fault-related growth strata in the lower Plan de los Yeuques Formation. Charrier et al. (1996) proposed that these structures were active as major basin-bounding normal faults during Paleogene extension and were later inverted during Miocene shortening, and this model has been subsequently developed by other studies (Charrier et al., 2002, 2024; Piquer et al., 2017). Mosolf et al. (2018) instead interpreted these faults as parts of a long-lived (ca. 37 Ma to ≤11 Ma) transpressive shear zone, citing dextral-reverse kinematic indicators and hanging-wall deposits characterized by slow accumulation rates and an absence of footwall-derived detritus. These interpretations are not entirely mutually exclusive and may reflect polyphase deformation. Resolving their crustal thickness implications requires independent geochemical constraints, which we develop below.

4. Zircon Petrochronology

Integrating U-Pb geochronology with trace element geochemistry (i.e., zircon petrochronology) is a powerful tool in reconstructing arc magmatic evolution, recording magma chemistry and crystallization conditions (e.g., Claiborne et al., 2018; Ferry & Watson, 2007; Loucks et al., 2020). Although several trace element ratios have been proposed as crustal thickness proxies (e.g., Balica et al., 2020; Tang et al., 2021), recent work shows complex dependence on melt source and evolution (e.g., Bell & Kirkpatrick, 2021; Roberts, Spencer, et al., 2024; Yakymchuk et al., 2023). We therefore evaluate multiple zircon parameters to assess both magmatic evolution and proxy fidelity.

4.1. Samples and Approach

We determined U-Pb ages and trace element compositions of 294 zircons from 22 samples of Cretaceous–Quaternary rock in the Tinguiririca area, including 21 primary tuffs and 1 volcanoclastic sandstone. These include 3 new samples and 19 collected by Mosolf (2013) and Mosolf et al. (2018). Tuff samples were collected from pyroclastic flows up to 30 m thick and containing lapilli and bombs, interpreted to have accumulated near volcanic vents (within tens of km). All samples yielded unimodal zircon age populations coherent with adjacent geochronologic constraints (see Section 4.2). Thus, we interpret the zircons in our samples to be mostly derived from local magmatism, with minimal inheritance or recycling.

Phenocryst assemblages of the analyzed samples provide context for interpreting zircon trace-element trends. All analyzed tuff samples with available petrographic information contain plagioclase phenocrysts (Data Set S1). Hornblende is present in a minority of Plan de los Yeuques and Abanico Formation samples, but absent in the Miocene unit, whereas biotite is ubiquitous in tuffs of the Miocene unit but rare to absent in older formations. The Quaternary sample also contains plagioclase and hornblende, and is the only sample with recognized pyroxene.

Zircons were analyzed by laser ablation (LA) sector-field inductively coupled plasma mass spectrometry (ICP-MS) at the Arizona LaserChron Center, following methods outlined by Balica et al. (2020). We filtered for unaltered igneous zircon by removing analyses exhibiting: (a) La > 1 ppm (probable mineral inclusions), (b) Th/U < 0.1 (metamorphic origin), (c) europium anomaly (Eu/Eu*; measured Eu relative to Sm and Gd) > 1 (feldspar inclusions), or (d) Dy/Nd + Dy/Sm < 60 (chemical alteration; e.g., Bell et al., 2019; Tang et al., 2024, and references therein). Zircons with elevated P (≥750 ppm)—potentially indicative of S-type magma sources (e.g., Burnham & Berry, 2017)—were flagged and excluded from crustal thickness proxy analysis to avoid potential contamination by crustal melts, but were retained for independent evaluation of melt source evolution.

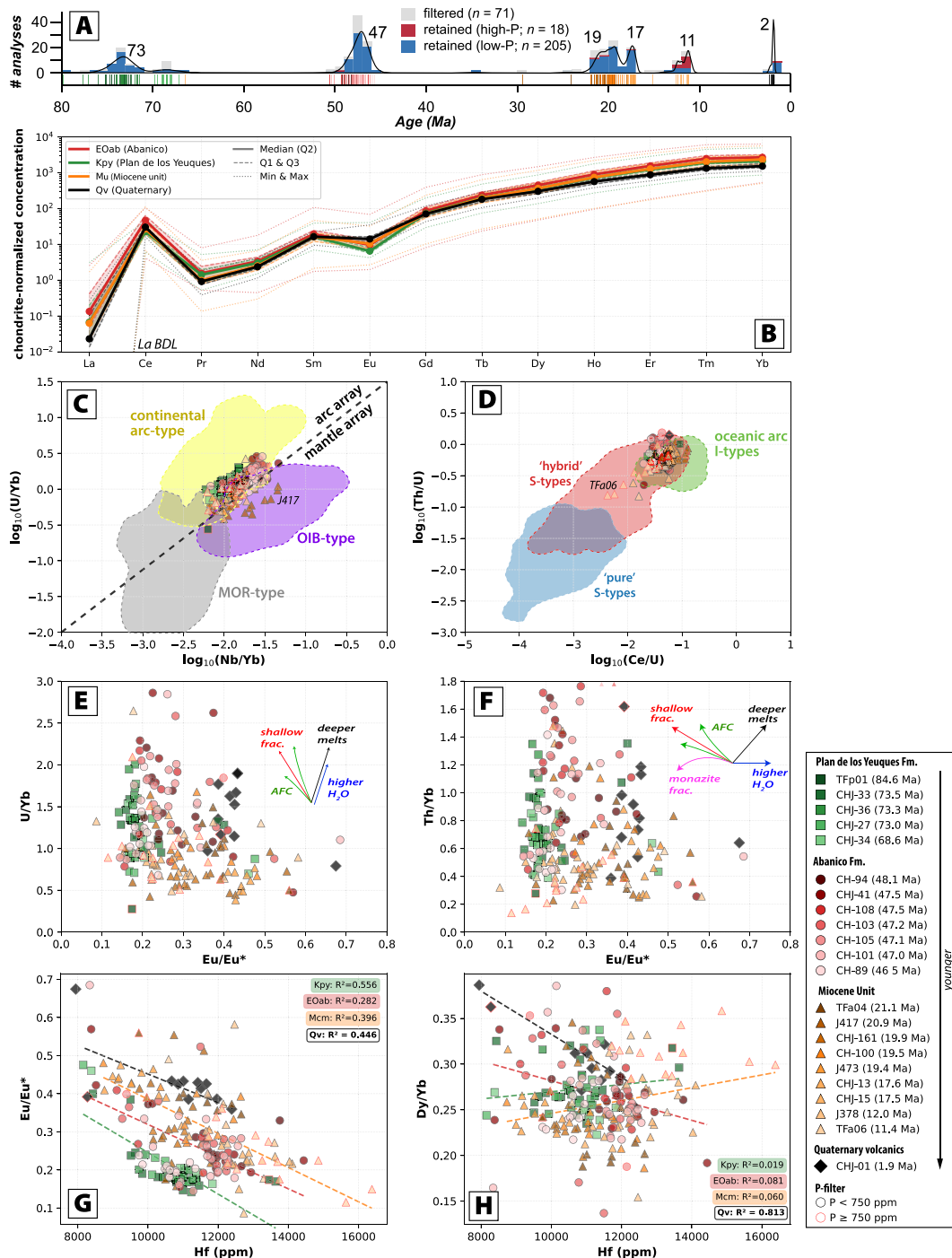


Figure 5. Tectonic discrimination and process interpretation diagrams for zircon petrochronologic data. (a) Probability density plot (PDP) and histogram (1 Myr bins) of zircon analyses. PDP includes only retained analyses. Bold numbers indicate programmatically picked age-probability peaks. Rug plot at bottom shows analyses colored by sample (legend at bottom right). (b) Chondrite-normalized zircon REE patterns, showing quartile ranges by geologic unit. (c) U/Yb versus Nb/Yb tectonic discrimination diagram (after Grimes et al., 2015) shows that most analyses plot within the continental arc field. (d) Th/U versus Ce/U discrimination diagram (after Roberts, Yakymchuk, et al., 2024) shows that most analyses plot within the overlap region between I-type and “hybrid” S-type magmas. (e) U/Yb versus Eu/Eu* and (f) Th/Yb versus Eu/Eu* plots (after Bell & Kirkpatrick, 2021) show vertical to steeply negative trends for Cretaceous–Eocene samples and shallower, positive trends for Miocene–Quaternary samples; colored arrows show predicted trends for various processes. (g–h) Comparison of potential crustal thickness proxies zircon Eu/Eu* (g) and Dy/Yb (h) against the melt evolution monitor Hf (e.g., Bell & Kirkpatrick, 2021; Claiborne et al., 2010). Linear regressions (dashed lines) are shown for each unit. Abbreviations: BDL—below detection limits; OIB—ocean island basalts; MOR—mid-ocean ridge; AFC—assimilation–fractional crystallization.

After filtering, 223 analyses were retained for magma source characterization, with 205 low-P analyses used for crustal thickness proxy assessment. Details on sample preparation and analysis are available in Text S2 in Supporting Information S1. Analytical settings and complete U-Pb and trace element data are reported in Data Set S2.

4.2. Results and Interpretation

Zircon U-Pb geochronology yielded ages mainly between 1.83 and 79.8 Ma, with modes of ca. 2, 11, 17, 19, 47, and 73 Ma, and two additional Jurassic ages likely representing xenocrysts or detrital grains (Figure 5a). The dominance of unimodal age populations in tuff samples, with ages matching their stratigraphic positions, confirms that analyzed zircons are predominantly comagmatic, rather than inherited or recycled detrital grains. Chondrite-normalized rare-earth element (REE) patterns show positive Ce and negative Eu anomalies superimposed on a positive slope in light-to-heavy REE abundance (Figure 5b). These patterns are typical of igneous zircon (e.g., Hoskin & Schaltegger, 2003) and thus support the filtering criteria used to reject altered or metamorphic grains.

4.2.1. Tectonic Discrimination and Process Interpretation

Most analyses overlap with continental arc-derived zircon on the U/Yb versus Nb/Yb tectonic discrimination diagram of Grimes et al. (2015, Figure 5c). On the source discrimination diagram of Roberts, Yakymchuk, et al. (2024), most analyses fall within the overlap zone between zircon crystallizing from oceanic I-type and “hybrid” S-type melts (Figure 5d). Several analyses from the Miocene unit deviate from the main data cluster on both plots, trending toward the ocean island basalt (OIB)-type and “pure” S-type end-members, respectively (Figures 5c and 5d). Similar trends are evident on the P versus Dy discrimination diagram of Ackerson et al. (2021, Figure S1a in Supporting Information S1). These results suggest protracted subduction-related magmatism, with potential melt source diversification during the Miocene.

On bivariate plots of zircon (U, Th)/Yb versus Eu/Eu*, Cretaceous–Eocene analyses define steeply negative trends, indicating shallow fractionation or assimilation–fractional crystallization (AFC; Figures 5e and 5f; Bell & Kirkpatrick, 2021). Miocene–Quaternary samples exhibit flatter, positive trends consistent with mixing with deeper melts and/or increasing H₂O. All units show a negative correlation between Eu/Eu* and Hf concentrations ($R^2 = 0.28–0.57$; Figure 5g). This is consistent with co-crystallization of Eu-rich phases like plagioclase, which is present in the phenocryst assemblage of all samples. By contrast, Dy/Yb shows little correlation with Hf concentrations in Cretaceous–Miocene zircon ($R^2 \leq 0.08$; Figure 5h), suggesting minimal modification of this ratio during zircon crystallization. Similar relationships exist for Sm/Yb and Gd/Yb ($R^2 < 0.11$; Figures S1b–S1c in Supporting Information S1). However, our single Quaternary sample shows a negative Dy/Yb–Hf correlation ($R^2 = 0.82$), potentially indicative of co-crystallization with middle rare-earth element (MREE)-sequestering phases like amphibole, which was observed in the phenocryst assemblage of this sample (Data Set S1).

4.2.2. Temporal Evolution of Magmatic Conditions and Source

Figure 6 shows the temporal evolution of zircon trace-element proxies for several magmatic characteristics. We show both sample-level medians capturing the overall temporal trend and individual analyses showing significant within-sample scatter likely related to magma-process diversity, such as crystal zoning and the assembly of multiple melt batches (e.g., Andersen et al., 2019; Claiborne et al., 2010, 2018; Lowery Claiborne et al., 2006). Ti concentration is a proxy for crystallization temperature and correlates with water content in arcs (Ferry & Watson, 2007; Nathwani et al., 2024). Temperatures were calculated assuming α_{SiO_2} and α_{TiO_2} of unity; unconstrained activities of these phases may introduce uncertainties of $\sim 60–70^\circ\text{C}$, with an additional uncertainty of $\sim 50^\circ\text{C}$ from pressure variations of ± 1 GPa (Ferry & Watson, 2007). Our samples generally range from ~ 3 to 10 ppm ($\sim 650–750^\circ\text{C}$), aligning with typical “wet” arc systems, although two samples from the Miocene unit show higher values consistent with drier melts (Figure 6b).

Most analyses ($n = 205/223$) plot below the $P = 750$ ppm threshold for S-type magmas, indicating predominantly I-type character (Burnham & Berry, 2017). Correspondingly, Th/U medians of most samples are between $\sim 0.8–1.4$, consistent with a mantle-derived source (Figures 6c and 6d; Roberts, Yakymchuk, et al., 2024). The two youngest samples from the Miocene unit (ca. 13–10 Ma) differ markedly; 12 analyses show $P > 750$ ppm and

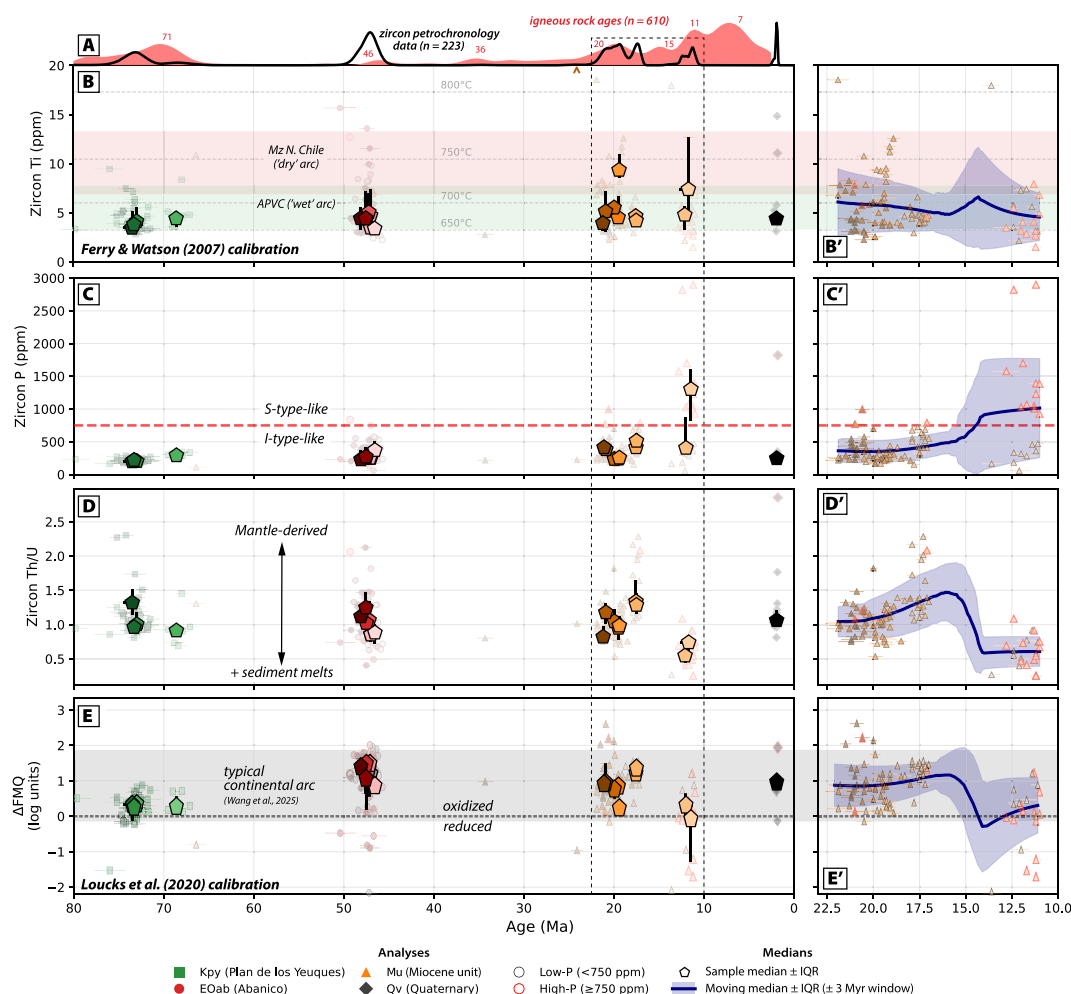


Figure 6. Temporal evolution of zircon trace element proxies for magmatic conditions. (a) PDP of retained zircon analyses from this study (black curve) and kernel density estimate (KDE) of compiled U-Pb and $^{40}\text{Ar}/^{39}\text{Ar}$ ages from igneous bedrock in the region of 33–36°S (red filled curve; 610 sample ages; 1 Myr kernel bandwidth; excludes Quaternary samples; see Text S4 in Supporting Information S1 for details), interpreted as a rough proxy for magmatic productivity (data from Mosolf et al., 2018; Muñoz et al., 2018, this study; Pilger, 2018, and references therein). Programmatically picked peaks are labeled for the KDE. Left panels (b–e) show full 80–0 Ma time series; filled symbols are sample-level medians, with error bars spanning the 1st–3rd quartile range; individual analyses are shown transparently. Right panels (b'–e') show the 23–10 Ma interval with a Gaussian-weighted moving median (navy line, ± 3 Myr window), and $\pm 1\sigma$ envelope (blue shading). (b, b') Ti concentration with temperature isotherms calculated using the thermometer of Ferry and Watson (2007), assuming α_{SiO_2} and α_{TiO_2} of unity. Comparative fields show value ranges from representative “dry” versus “wet” arc zircons, from Nathwani et al. (2024, and references therein); Mz N. Chile—Mesozoic arc of northern Chile; APVC—Altiplano-Puna Volcanic Complex. (c, c') P concentration with I/S-type threshold (750 ppm; Burnham & Berry, 2017). (d, d') Th/U ratio, a proxy for mantle- versus sediment-derived melt (Roberts, Yakymchuk, et al., 2024). (e, e') Oxygen fugacity with typical continental arc field (Loucks et al., 2020; Wang et al., 2025).

lower Th/U (Figures 6c and 6d). This P-enrichment is consistent with crystallization from melts generated from metasedimentary or mafic lower crust during apatite dissolution (Burnham & Berry, 2017), whereas lower Th/U may indicate crystallization of high-Th accessory phases in evolved crustal melts (Roberts, Yakymchuk, et al., 2024).

Magmatic oxidation state, expressed as oxygen fugacity ($f\text{O}_2$) relative to the fayalite–magnetite–quartz (FMQ) buffer, was calculated from zircon Ce–U–Ti systematics following Loucks et al. (2020). Calculated median $f\text{O}_2$ values are mainly +0.5 to +1.75 ΔFMQ , indicating moderately to strongly oxidized magmas typical of continental arc settings (Figure 6e). However, several samples from the Plan de los Yeuques and Miocene units show median values of $< +0.5$ ΔFMQ . In the context of melt source proxies above (Figures 6b–6d), this may reflect

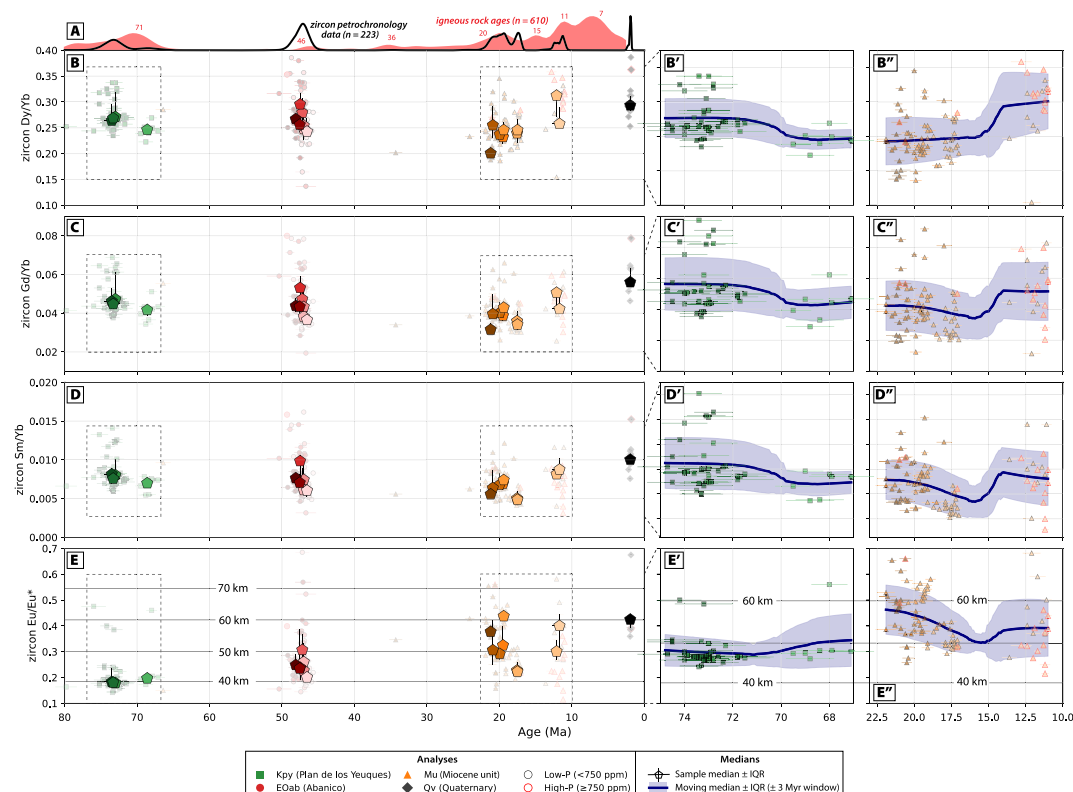


Figure 7. Temporal evolution of zircon trace element ratios interpreted as crustal thickness proxies. (a) PDP of retained zircon analyses from this study (black curve) and KDE of compiled igneous bedrock ages at 33–36°S (as in Figure 6a). (b–e) Full time series (80–0 Ma) showing (b) Dy/Yb, (c) Gd/Yb, (d) Sm/Yb, and (e) Eu/Eu* versus age; filled symbols are sample-level medians, with error bars spanning the 1st–3rd quartile range; individual analyses are shown transparently. Dashed boxes highlight zoomed intervals shown in adjacent panels. (b'–e') Late Cretaceous interval (75–67 Ma) and (b''–e'') Miocene interval (23–10 Ma), showing individual analyses with 1σ age uncertainties and Gaussian-weighted moving median (navy line, ±3 Myr window) with ±1σ envelope (blue shading). High-P zircons ($P \geq 750$ ppm; red outlines) are excluded from sample/ moving median calculations. Horizontal lines in (e) show crustal thickness estimates derived from the calibration of Tang et al. (2021), which are likely overestimates as discussed in the text.

greater input from relatively reduced mantle during the Late Cretaceous and from lower crustal lithologies during the Miocene.

4.2.3. Zircon Crustal Thickness Proxies

Informed by the preceding analysis of magma source and evolution, we evaluate zircon trace element ratios previously proposed as proxies for crustal thickness (e.g., Balica et al., 2020; Tang et al., 2021). Zircon (Sm, Gd, Dy)/Yb values show broadly consistent long-term trends, including a general decrease during deposition of the Plan de los Yeuques Formation (ca. 75–66 Ma) and scattered values for the Abanico Formation, with similar sample medians to the Plan de los Yeuques Formation (Figures 7b–7d). In Miocene unit samples, Dy/Yb remains stable from ca. 22 to 17 Ma, whereas Sm/Yb and Gd/Yb instead show a transient decrease over the same interval. All three ratios rise in concert between ca. 17 and 13 Ma (Figures 7b–7d). Quaternary zircon yields Dy/Yb values similar to those in the upper Miocene unit (Figure 7b), whereas Sm/Yb and Gd/Yb are elevated (Figures 7c and 7d). This divergence may reflect preferential depletion of Dy relative to lighter REE during amphibole fractionation, consistent with the Dy/Yb–Hf correlation and presence of amphibole phenocrysts in the Quaternary sample (Figure 5h; Data Set S1).

Zircon Eu/Eu* displays a contrasting trend. It generally increases from ca. 73 Ma to ca. 20 Ma, decreases until ca. 17 Ma, then rises to ~0.4 in Middle Miocene–Quaternary zircon (Figure 7e). The Tang et al. (2021) Eu/Eu* calibration yields a Quaternary crustal thickness estimate of 60.34 ± 7.23 km, which exceeds modern geophysical constraints by >10 km (Figure 7e).

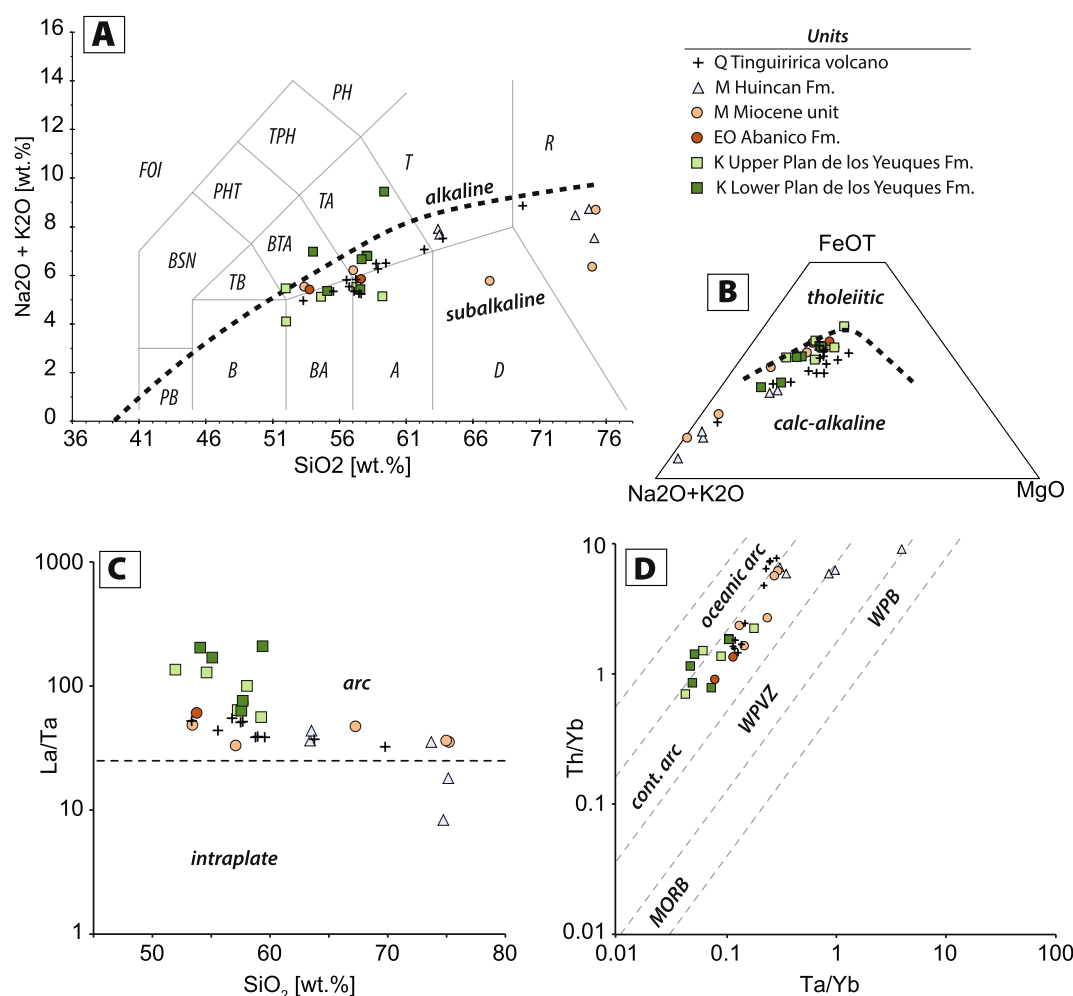


Figure 8. Whole-rock geochemical classification (a–b) and tectonic discrimination (c–d) diagrams. Plotted samples are those retained after the major-element and alkalinity filters described in Section 5.1; the dashed curve in (a) shows the alkaline–subalkaline boundary used in filtering. Most samples plot in the subalkaline and calc-alkaline fields on (a) total alkali silica (TAS) and (b) alkalis, total iron, magnesium (AFM) diagrams, respectively. Abbreviations in (a) are from Le Maitre (2002). All Tinguiririca samples fall within arc fields in (c) La/Ta versus SiO₂ and (d) Th/Yb versus Ta/Yb tectonic discrimination diagrams, although 2 (c) to 3 (d) Valle Hermoso samples plot within intraplate fields.

5. Whole-Rock Geochemistry and Paleomohometry

We combine our zircon petrochronology with whole-rock paleomohometry, based on empirical correlations between arc magma geochemistry and sub-arc Moho depth (Luffi & Ducea, 2022b). This approach facilitates proxy comparison and provides quantitative estimates of crustal thickness.

5.1. Samples and Approach

We collected 13 volcanic rock samples from the Plan de los Yeuques Formation, Abanico Formation, and Miocene unit in the Tinguiririca area (Figures 2 and 3). Major oxides were analyzed by X-ray fluorescence at the Hamilton Analytical Laboratory at Hamilton College, and trace elements were analyzed by LA-ICP-MS at the Colorado School of Mines (see Text S3 in Supporting Information S1; analytical data in Data Set S3). Additionally, we compiled published data from 22 samples reported from the Plan de los Yeuques Formation (Muñoz et al., 2018), the Middle–Late Miocene Huincan Formation in Valle Hermoso (location in Figure 1, ~25 km east of Termas del Flaco; Sruoga et al., 2020), and Quaternary Tinguiririca volcano lavas (Jacques et al., 2013, Data Set S4 in Supporting Information S1).

Before analysis, we filtered out samples with major element totals outside 95–101 wt.%. Major elements were then normalized to a dry 100 wt.% sum. In accordance with crustal thickness calibrations of Luffi and Ducea (2022b), we considered only analyses with 45–80 wt.% SiO₂ and total alkali contents within 1.5 wt.% of the alkaline-subalkaline boundary at a given wt.% SiO₂. These filters rejected nine samples.

Following evaluation using geochemical classification and tectonic discrimination diagrams (Figure 8), we used these data to reconstruct sub-arc crustal thicknesses via the Geochemical Arc Moho Estimator (GAME) MATLAB program of Luffi and Ducea (2022b). This model uses up to 41 geochemical parameters (i.e., sensors) empirically calibrated against Pliocene–Quaternary arc segments with well-constrained crustal thickness. For each stratigraphic unit, samples were grouped into MgO bins, then the bin-averaged value of the best-performing sensors (selected by a programmatic, iterative workflow, see Luffi & Ducea, 2022b) was used to calculate a median crustal thickness and uncertainty, represented by the median absolute deviation (MAD).

We intentionally restricted the input data set to samples from the Tinguiririca section and immediately adjacent areas of the Eastern Principal Cordillera (within ~25 km; samples described above; Figure 1) to minimize spatial confounds and enable comparison to local geologic constraints. However, this resulted in a relatively small number of retained samples per stratigraphic unit. As a consequence, GAME failed to produce estimates for the Abanico Formation and Quaternary Tinguiririca volcano using the default sensor filters due to insufficient MgO bin coverage. As a workaround, we programmatically relaxed the filters to maximize the amount of data used for each crustal thickness estimate (see Text S3 in Supporting Information S1 for filter parameters). Broadening the filtering criteria included sensors that would otherwise be omitted due to high scatter or limited MgO bin representation, increasing total measurements per estimate at the cost of greater uncertainty. Using this approach, our reported crustal thickness estimates are each based on 55–187 geochemical measurements ($n_{\text{retained samples}} \times n_{\text{unique sensors}}$; Table S1 in Supporting Information S1). Detailed results of our GAME models are reported in Figures S2–S7 in Supporting Information S1 and Data Set S4.

5.2. Results and Interpretation

Retained samples exhibit SiO₂ values in the range of 52–75 wt.% (Figure 8a). Most samples exhibit sub-alkaline character and calc-alkaline affinity (Figures 8a and 8b) and plot within the arc field on tectonic discrimination diagrams. Two to three Miocene samples from the Huincan Formation plot within adjacent portions of the intraplate field (Figures 8c and 8d). Overall, these results align with zircon petrochronologic interpretations to suggest our data set primarily reflects magmatism in an arc setting.

Median crustal thickness estimates decrease from 37.1 ± 9.2 km to 33.8 ± 5.4 km from the lower (ca. 80–75 Ma) to upper (ca. 75–63 Ma) Plan de los Yeuques Formation, although uncertainties are large and overlapping (Figure 9a). Samples from the Abanico Formation (ca. 52–37 Ma) and Miocene unit (ca. 20–16 Ma) yield similar estimates of 35.5 ± 3.9 km and 35.6 ± 2.6 km, respectively. The Huincan Formation yields a median value of 44.1 ± 2.4 km, suggesting 8.5 ± 3.6 km of crustal thickening during Miocene time (uncertainty propagated in quadrature; Figure 9a). We note the caveat that Huincan Formation samples are reported from 25 km east of the main study area. Quaternary samples from the Tinguiririca volcano yield an estimate of 46.0 ± 5.7 km, overlapping with independent constraints on modern crustal thickness below the study area of 44 ± 2 km from isostasy (Luffi & Ducea, 2022b); 45.5 ± 2 km from the CRUST 1.0 model (Laske et al., 2013); and 46.5 ± 2 km from gravity data (Tassara & Echaurren, 2012). Relative to the Miocene unit (35.6 ± 2.6 km), the Quaternary GAME estimate indicates 10.4 ± 6.3 km of total crustal thickening in the study area since ca. 18 Ma (Figure 9).

6. Discussion

6.1. Viability of Zircon Geochemical Crustal Thickness Proxies

Before discussing tectonic implications, we evaluate zircon crustal thickness proxies by comparing them to whole-rock paleomohometry and geologic constraints. Zircon Eu/Eu* yields temporal trends inconsistent with whole-rock paleomohometry (Figures 7 and 9) and overestimates Quaternary crustal thickness by >10 km relative to geophysical constraints (using the calibration of Tang et al., 2021). Zircon Eu/Eu* trends are also inconsistent with expectations from geologic observations, in that they increase during documented Late Cretaceous normal faulting (Figures 4a, 4b, and 7e; Muñoz et al., 2018) yet decrease during well-resolved

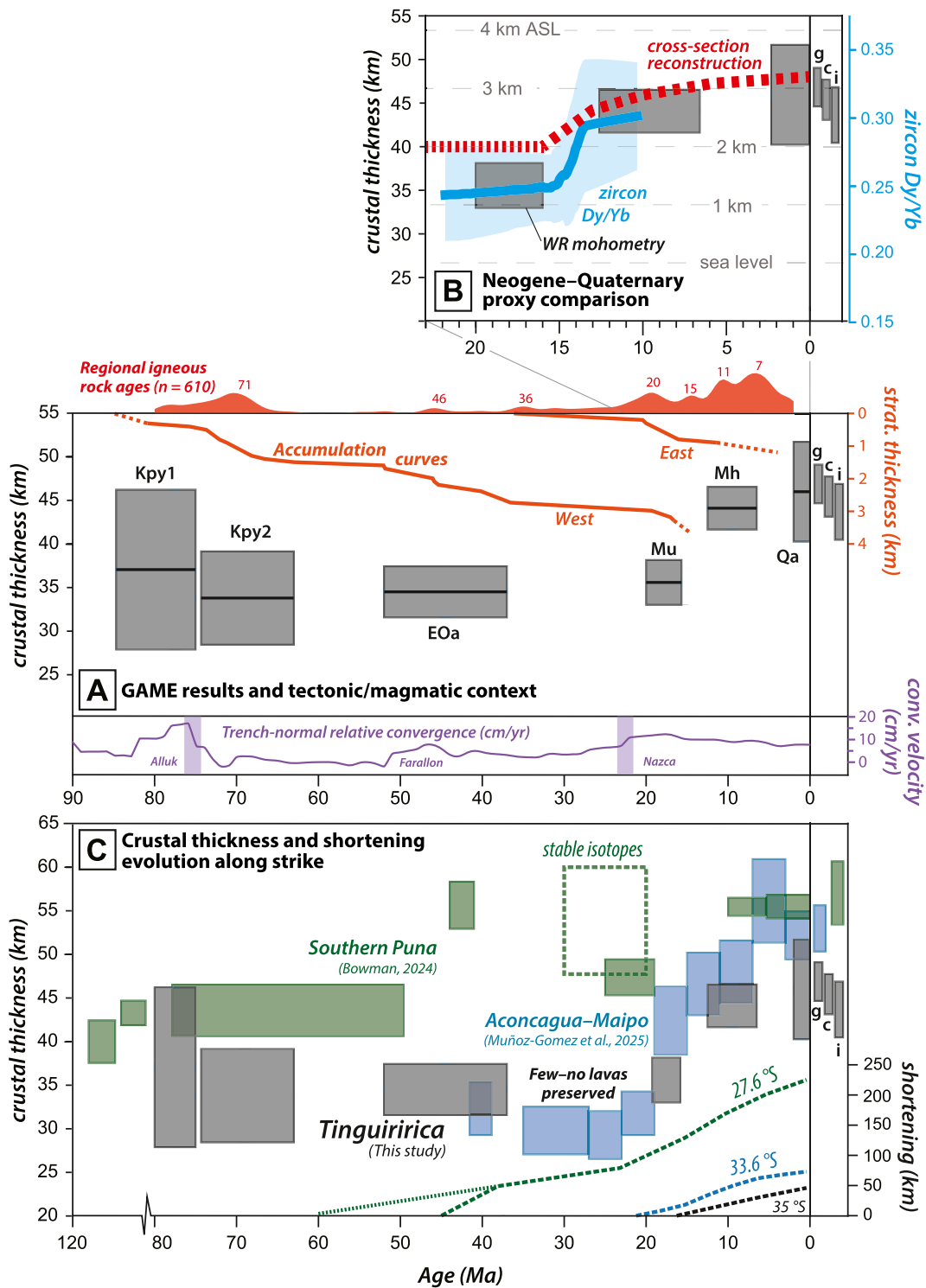


Figure 9.

Miocene crustal shortening (Figure 7e; e.g., Fuentes et al., 2016; Giambiagi et al., 2022; Horton et al., 2016; Howlett, 2025).

Temporal variations in calculated fO_2 closely track those of Eu/Eu^* (Figures 6e and 7e), suggesting that shifts in magmatic oxidation state exert a dominant control. Phase equilibrium modeling by Yakymchuk and Kirkland (2025) supports this interpretation, demonstrating that source Fe^{3+}/Fe^{2+} ratios can cause zircon Eu/Eu^* to vary by >0.2 at a given pressure. Furthermore, the low zircon Ti concentrations in our data set (~ 3 – 10 ppm; Figure 6b) indicate wet magmatic conditions (Nathwani et al., 2024), which suppress plagioclase crystallization in the melt source region and would further elevate Eu/Eu^* independent of crustal thickness. Additional modification by shallow fractionation and AFC is likely indicated by (U, Th)/Yb correlations (Figures 5e and 5f). Overall, we conclude that zircon Eu/Eu^* in our data set was modified by multiple factors decoupled from crustal thickness—including source redox, water content, and shallow crustal effects, and its utility as a crustal thickness sensor is thus limited.

These effects may be more pronounced in studies assessing zircon geochemistry within a narrow arc segment (e.g., this study; Bell & Kirkpatrick, 2021). At the broader batholith or orogen scale, Eu/Eu^* has shown reasonable agreement with independent constraints where averaging across heterogeneous populations reduces the influence of local redox variations and AFC processes (e.g., Sundell et al., 2024; Tang et al., 2024). Nevertheless, zircon Eu/Eu^* crustal thickness estimates in the Andes exceed geophysical constraints even in these large-scale studies (e.g., Carrapa et al., 2022; Sundell et al., 2022). This may reflect secular or regional differences from the calibration data set, such as the relatively wet character of Andean arc magmas (Nathwani et al., 2024).

By contrast, zircon (Sm, Gd, Dy)/Yb ratios broadly track temporal trends in whole-rock mohometers and align with geologic observations, decreasing during Late Cretaceous normal faulting and increasing during Miocene contractional growth strata deposition (Figures 4, 7, and 9; Mosolf et al., 2018; Muñoz et al., 2018). These proxies show general long-term mutual agreement, although they diverge somewhat during the Neogene–Quaternary (Figures 7b–7d). Luffi and Ducea (2022b) demonstrated that whole-rock (Sm, Gd, Dy)/Yb ratios show minimal covariation with MgO in global arc samples, interpreted to suggest that their correlation with sub-arc Moho depth primarily reflects crustal thickness-modulated mantle melting degree rather than intracrustal processes (e.g., Turner et al., 2016). Zircon (Sm, Gd, Dy)/Yb ratios similarly show minimal covariation with Hf concentrations for most units (Figure 5h), suggesting analogous preservation of melt-source signatures that are sensitive to crustal thickness. An exception is Quaternary Dy/Yb, for which Hf correlation ($R^2 = 0.82$) may record cogenetic amphibole crystallization (Figure 5h). However, this does not compromise our first-order Miocene thickening interpretation, as all three ratios register a significant increase in ca. 17–13 Ma samples, which lack amphibole (Figure 7; Data Set S1).

Zircon (Sm, Gd, Dy)/Yb ratios offer an analytical advantage over lighter REE-based proxies (e.g., La/Yb) in that the numerator elements occur in zircon at concentrations well above LA-ICP-MS detection limits (e.g., Burnham, 2020). Of these, Dy is present in the greatest quantities in zircon [99.3 ± 28.6 ppm (median $\pm$ MAD) in our data set], and Dy/Yb shows a trend most similar to that of whole-rock paleomohometry results (Figure 9b). We therefore view these ratios, and especially Dy/Yb, as the most promising crustal thickness proxies currently available in zircon—provided that possible modification by shallow fractionation of REE-sequestering

Figure 9. (a) Crustal thickness estimates calculated using whole-rock paleomohometry (GAME). Boxes show unit-level pooled estimates (median $\pm$ MAD) for discrete units. Modern crustal thickness estimates (bars at right) are from: g—gravity data; c—CRUST 1.0, and i—isostasy (Laske et al., 2013; Luffi & Ducea, 2022b; Tassara & Echaurren, 2012). Unit abbreviations: Kpy1/2—Lower/Upper Plan de los Yeuques Fm.; EOab—Abanico Fm.; Mu—Miocene unit; Mh—Huincan Fm.; Qa—Tinguiririca volcano. Red curve at top shows KDE of compiled igneous bedrock ages at 33–36°S (as in Figure 6a). Orange curves show cumulative stratigraphic (strat.) thickness through time for Tinguiririca west and east sections, uncorrected for compaction (upper right axis; from Figure 3c). Purple curve at bottom shows trench-normal relative convergence (conv.) velocity between South America and the Aluk/Farallon/Nazca plates at 35°S (lower right axis), from the plate reconstruction of Torsvik et al. (2019, see Text S5 in Supporting Information S1). Vertical bars show timing of plate reorganization. (b) Neogene–Quaternary crustal thickness estimates from GAME (boxes, left axis) compared to those from area-balanced cross-section reconstructions (dashed red line, left axis; Giambiagi et al., 2022). Zircon Dy/Yb from Figure 7b is shown for trend comparison (blue line and shading; right axis). Dashed lines show arc elevation estimates assuming isostatic equilibrium, following Luffi and Ducea (2022b). (c) Along-strike comparison of paleomohometry results (boxes, left axis; median $\pm$ MAD) from the Tinguiririca area ($\sim 35^\circ$ S; this study), the southern Puna (~ 24.5 – 27.5° S; Bowman, 2024), and Aconcagua–Maipo (~ 32.5 – 34.5° S; Muñoz-Gómez et al., 2025) with cumulative shortening from balanced cross sections in each segment (curves, right axis; Giambiagi et al., 2022). Dashed green box shows crustal thickness inferred from stable isotope-based elevation estimates of Quade et al. (2015). Short dashed green line shows foreland basin migration rate at ~ 23 – 26° S from DeCelles et al. (2011), suggesting earlier shortening at the southern Puna latitude.

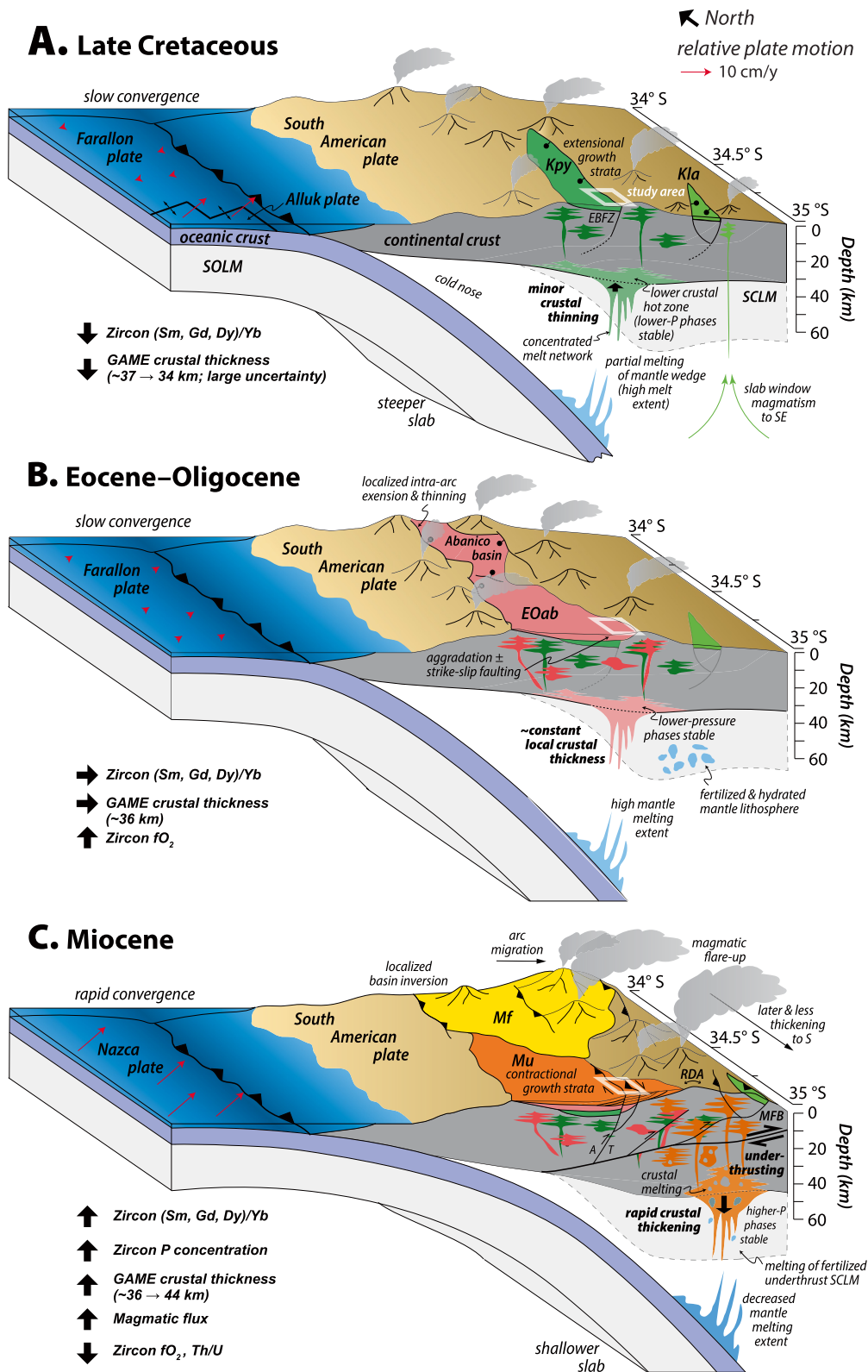


Figure 10.

phases is taken into account. Evaluation of these ratios against melt evolution monitors such as Hf (e.g., Bell & Kirkpatrick, 2021; Claiborne et al., 2010) may aid in this, providing a means to assess modification by cogenetic phases. At present, these proxies lack quantitative crustal thickness calibration and are best applied qualitatively alongside whole-rock data to provide enhanced temporal resolution.

6.2. Crustal Thickness Evolution of the Southern Central Andes

Informed by the preceding proxy assessment, we evaluate models for tectonic development of the orogenic segment in the context of our results and a synthesis of regional structural and geochemical data.

6.2.1. Late Cretaceous Crustal Thinning and Extension

Whole-rock paleomohometry and zircon (Sm, Gd, Dy)/Yb results indicate modest Late Cretaceous crustal thinning during deposition of the Plan de los Yeuques Formation, consistent with syndepositional normal faulting in the study area (Figures 4a, 4b, 7c–7e, and 9; Muñoz et al., 2018; Encinas et al., 2024) and previous interpretation of extensional deposition of the partially correlative Colimapu Formation and Los Angeles Unit (Henríquez et al., 2024; Iannelli et al., 2018). Plate reconstructions indicate that initial deposition of the Plan de los Yeuques Formation occurred soon after a major plate reorganization (ca. 85–83 Ma), triggering decreased trench-normal relative convergence and increased obliquity along the Farallon–South America margin (Figure 9a; Torsvik et al., 2019). This temporal coincidence suggests that crustal thinning and normal faulting may have been driven primarily by slow, highly oblique convergence during initial Farallon subduction (Figure 10a; e.g., Muñoz et al., 2018).

Previous workers attributed extension in the Los Angeles Unit (~120 km southeast) to slab window formation during Aluk–Farallon spreading ridge subduction (Figure 10a; Iannelli et al., 2018). However, the Plan de los Yeuques Formation lacks enriched asthenospheric components expected in a slab window scenario (Figures 5c, 5d, 8c, and 8d; Muñoz et al., 2018), suggesting minimal direct influence by ridge subduction processes in the Tinguiririca area. The lack of clear slab window magmatism in the Tinguiririca area suggests either that ridge subduction initiated south of the study area, or that a transform segment subducted below the region, consistent with significant segmentation of the Aluk–Farallon ridge inferred elsewhere by Gianni et al. (2019).

The extent of orogenic growth during Cenomanian–Santonian orogenesis preceding deposition of the Plan de los Yeuques Formation is debated. Several workers suggest development of significant fold-thrust belt topography (e.g., Balgord & Carrapa, 2016; Fennell et al., 2017; Tunik et al., 2010), whereas others infer only localized compression (Encinas et al., 2024). Our lowermost Plan de los Yeuques Formation estimate (37.1 ± 9.2 km) provides an indirect constraint, as the formation conformably overlies the BRCU (Muñoz et al., 2018). Substantially thickened crust during BRCU deposition would require a rapid, unrecorded thinning event for which no structural evidence exists. Therefore, our results suggest that the crust at ~35°S was at most modestly thickened prior to the extensional phase recorded by the Plan de los Yeuques Formation. Independent geochemical data from the Las Chilcas Formation at ~33°S similarly indicate thin Late Cretaceous crust (Gerding et al., 2023).

6.2.2. Implications for Paleogene Intra-Arc Rift Models

During Paleogene time, the Abanico basin and correlative depocenters are thought to record a second, more widespread episode of intra-arc extension (e.g., Charrier et al., 2005, 2024; Fernández Paz et al., 2024, and references therein). This interpretation is supported by normal fault-related growth strata that have been reported

Figure 10. Block diagrams showing tectono–magmatic evolution of the southern Central Andes at ~34–35°S. (a) Late Cretaceous: Modest crustal attenuation and normal faulting within the El Baule fault zone (EBFZ) is recorded by the Plan de los Yeuques Formation (Kpy), coinciding with slow convergence between the Farallon and South American plates and slab-window magmatism recorded by the Los Angeles Unit (Kla) to the southeast (e.g., Iannelli et al., 2018). (b) Eocene–Oligocene: The Abanico Formation (EOab) is accommodated by localized normal faulting and aggradation within an intra-arc basin. Crustal thickness remains roughly constant in the study area, indicating spatially distributed rather than focused extension. Note that the Abanico Formation ranges into the Lower Miocene in northern areas. (c) Miocene: Rapid convergence drives eastward arc migration, shortening within the Malargüe fold-thrust belt (MFB), and rapid crustal thickening (~10 km) by foreland underthrusting. Thickening initiates earlier and is greater in magnitude to the north. The Miocene unit (Mu) preserves growth strata related to uplift of the Río Damas anticline (RDA). Normal faults invert in northern portions of the Abanico basin, followed by deposition of the Farellones Formation (Mf). Later transpressional activation of the El Baule and El Fierro fault systems deforms the Tinguiririca section. Enhanced crustal melting and magmatism follow thickening onset. Red arrows show plate velocities relative to fixed South America at (a) 80 Ma, (b) 35 Ma, and (c) 18 Ma, from Torsvik et al. (2019).

elsewhere in the Abanico Formation (e.g., Jara & Charrier, 2014; Piquer et al., 2017) and prior geochemical paleomohometry results north of the study area suggesting crustal thinning between ca. 40 and 23 Ma (Muñoz-Gómez et al., 2025). The El Baule and/or El Fierro fault zones have been proposed to form part of a major normal fault system (the El Diablo fault zone) bounding the eastern basin margin for >1,100 km along strike and accommodating most of this extension (Charrier et al., 2024).

Our results, however, show no evidence for net crustal thinning over this interval. Whole-rock paleomohometry models yield similar crustal thicknesses for the Late Cretaceous (33.8 ± 5.4 km), Paleogene (35.5 ± 3.9 km), and Early Miocene (35.6 ± 2.6 km; Figure 9). Although temporal gaps encompass much of the proposed rifting stage, Paleogene (ca. 37 Ma) and Early Miocene (ca. 20 Ma) samples bracketing this interval show no change in crustal thickness. Zircon (Sm, Gd, Dy)/Yb ratios also maintain generally consistent median values throughout (Figures 7b–7d). Although the transient decrease in (Sm, Gd)/Yb between ca. 22 and 17 Ma could, in principle, indicate minor Early Miocene crustal thinning, this decrease is absent in Dy/Yb and does not correlate with local evidence for normal faulting (Figures 7b–7d). Overall, our data do not provide strong evidence for net crustal attenuation during Paleogene time, although transient thinning and rethickening cannot be excluded. An additional possibility is that magmatic addition at the base of the crust or mid/lower crustal flow compensated for any extensional attenuation, maintaining net crustal thickness even during active rifting (e.g., Fountain, 1989; Haschke & Günther, 2003).

Stratigraphic evidence in the study area further constrains the role of the El Baule and El Fierro fault systems during this interval. Revised sediment accumulation curves incorporating our new zircon U-Pb analyses show limited deposition during the proposed main rifting stage (ca. 36–18 Ma; Figure 3c), consistent with the observation of Mosolf et al. (2018) that low accumulation rates indicate minimal accommodation in the hanging walls of these faults. Combined with our crustal thickness results above, these observations indicate that the El Baule and El Fierro structures were either inactive or accommodated mainly strike-slip deformation during the Paleogene, rather than accommodating significant shortening or extension. This interpretation contrasts with earlier interpretations of the El Fierro Fault System as an originally basin-bounding normal fault later inverted during the Miocene (e.g., Charrier et al., 1996, 2002) but aligns with recent structural and geochronologic work along strike between ~32 and 36°S, which reinterprets other segments of the proposed El Diablo fault zone as angular or buttress unconformities (Encinas et al., 2024; Fennell et al., 2023; Howlett, 2025; Peluffo et al., 2026; Piñeiro et al., 2025; Ronemus, 2025). Combined with our local data, these reinterpretations indicate that this fault system did not operate as a through-going normal fault bounding the Abanico basin along its proposed length (cf. Charrier et al., 2024).

Our results do not preclude extension and inversion at the principal Abanico depocenter, but require its bounding structures to lie west of our study area, where thicker accumulations of the Abanico and correlative formations are mapped (Figure 1; e.g., Contreras & Schilling, 2021; Piquer et al., 2017) and prior igneous geochemistry suggests Paleogene crustal thinning (e.g., Fernández Paz et al., 2024; Muñoz-Gómez et al., 2025; Quiñones et al., 2023; Vergara et al., 1999). At our latitude, these results constrain only the eastern margin of the Abanico basin, which appears to have accumulated atop a buttress unconformity in the absence of local extension, rather than within an actively subsiding graben (Figure 10b).

Our Paleogene crustal thickness estimates also constrain models proposing Paleogene shortening at this latitude. Mosolf et al. (2018) interpreted deposits spanning ca. 37–11 Ma as syn-compressional based on inferred growth geometries, but our geochemical data record no detectable crustal thickening until ca. 17 Ma. The angular unconformity below the Abanico Formation east of the El Baule fault zone—which cuts progressively through the BRCU and into the Baños del Flaco Formation southward (Waite, 2005)—may reflect footwall erosion during Late Cretaceous extension or localized uplift during Paleogene strike-slip faulting, rather than significant Paleogene crustal shortening (Mosolf et al., 2018).

6.2.3. Miocene Thickening: Timing, Mechanism(s), and Proxy Agreement

Geochemical results from the Miocene unit and Huincan Formation indicate crustal thickening of 8.5 ± 3.6 km during Miocene time (Figure 9a; uncertainty propagated in quadrature). The Huincan Formation samples are reported from ~25 km east of the main Tinguiririca study area (Figure 1), so the whole-rock thickening signal could in principle reflect eastward migration of the magmatic arc into a previously thicker crustal column. However, the concurrent increase in zircon (Sm, Gd, Dy)/Yb ratios is recorded within the Tinguiririca section

itself (Figures 7b–7d) and therefore suggests in situ crustal thickening. Zircon petrochronology brackets the most rapid thickening between ca. 17 and 13 Ma (Figure 7), lagging by ~6 Myr an increase in trench-normal convergence rates from <5 to >10 cm/yr at ca. 23 Ma (Figure 9a; Torsvik et al., 2019). This thickening closely followed the onset of retroarc shortening in the Malargüe fold-thrust belt, evidenced by: (a) growth geometries in the Miocene unit recording syndepositional uplift of the Río Damas anticline (Figure 4c; Mosolf et al., 2018); (b) thermochronologic data indicating rapid cooling of western fold-thrust belt structures after ca. 20 Ma (Bande et al., 2020; Howlett, 2025); and (c) foreland basin deposits documenting rapid flexural subsidence and arrival of thrust-belt-derived detritus after ca. 19 Ma (e.g., Horton et al., 2016).

We infer that the Neogene onset of rapid retroarc shortening was the primary trigger for the main phase of crustal thickening below the WPC. This interpretation is consistent with seismic imaging in this segment, indicating a west-dipping crustal-scale detachment underthrusting foreland lithosphere below the arc (Ammirati et al., 2022). Area-balanced cross-section reconstructions of Giambiagi et al. (2022) resolve ~8 km of Miocene–Quaternary hinterland crustal thickening via this process, within uncertainty of our geochemical estimate of 10.4 ± 6.3 km (Figure 9b). However, achieving this thickening in their model requires 46 ± 8 km of retroarc shortening, which exceeds estimates from cross sections at similar latitude by other authors (≤ 28 km; Fuentes et al., 2016; Howlett, 2025).

This discrepancy may suggest that processes beyond foreland underthrusting contributed to the crustal thickness budget. One possible contributor is lower crustal flow, which has been invoked elsewhere in the Andes (Carrapa et al., 2022; Ouimet & Cook, 2010) but cannot be evaluated with available data and is not considered further. An additional potential mechanism is underplating of tectonically eroded material from the forearc. Stern (2020) estimated that ~85 km of forearc crust was subducted along this segment during the Miocene, and isotopic signatures of Miocene volcanics have been interpreted to indicate accumulation of this material below the hinterland (Kay et al., 2005). Although we cannot directly test this contribution with our data, the volume of forearc material removed could account for a meaningful fraction of the unresolved thickening budget.

A final potential thickening mechanism we consider is magmatic addition (e.g., Haschke & Günther, 2003). While mass addition is difficult to estimate directly, a regional compilation of igneous rock ages from 33 to 36°S (610 ages; see Text S4 in Supporting Information S1) provides a productivity proxy and shows increasing age frequency throughout the Miocene, peaking at ca. 7 Ma (Figures 6a and 7a; Pilger, 2018). This proxy is subject to lithologic bias, in that more mafic and altered Abanico Formation lithologies may yield fewer datable zircons and reliable $^{40}\text{Ar}/^{39}\text{Ar}$ ages compared to Miocene units (e.g., Cawood et al., 2012; Montecinos et al., 2008). However, the generalized volcanostratigraphy of Piquer et al. (2017) near the study latitude corroborates the apparent trend, showing peak accumulation rates of ~670 m/Myr during the Middle Miocene (>250 m/Myr above neighboring intervals), although stratigraphic thicknesses and age control in that compilation carry significant uncertainty. This is further supported by the regional Cenozoic intrusive record, which is characterized almost entirely by Neogene plutons on both sides of the divide at 33–36°S (e.g., Contreras & Schilling, 2021; Deckart et al., 2010; Makshev et al., 2009; Mescua et al., 2014; Nullo et al., 2005; Stern et al., 2011; Sruoga et al., 2005), although this could partly reflect enhanced crustal magma storage under the compressional tectonic regime, rather than increased magmatic flux. Because the age-frequency, accumulation-rate, and intrusive records are subject to largely independent biases yet register the same Neogene increase, we interpret these observations to suggest that magmatic productivity generally increased in concert with the rate of retroarc shortening but peaked ~7 Myr after the phase of most rapid thickening.

In view of this temporal lag, the regional trend of magmatism aligns with models in which progressive underthrusting of melt-fertile foreland lithosphere enhances arc productivity (e.g., DeCelles et al., 2009; Ducea & Barton, 2007). Petrochronologic evidence supports enhanced melting of underthrust material after ca. 14 Ma when elevated zircon P concentrations (≥ 750 ppm) accompanied by low Th/U and $f\text{O}_2$ signatures (Figures 6c–6e) suggest increased contribution from reduced lower crustal lithologies. Correspondingly, a coeval decrease in zircon $\epsilon\text{Hf}(i)$ (initial Hf isotopic composition) from $\geq +7$ to ≤ -10 over the same interval in regional data sets (~32.5–34.5°S) has been interpreted to record melting of Proterozoic basement (Muñoz-Gómez et al., 2025). By this mechanism, underthrust foreland lithosphere would have contributed to crustal thickening both directly via shortening and indirectly through magmatic addition driven by enhanced arc productivity. Collectively, these structural, geochronologic, and geochemical data support a scenario in which increased plate convergence promoted rapid retroarc shortening, driving local crustal thickening and magmatic flare-up (Figure 10c).

6.3. Along-Strike Variations and Implications for Andean Orogenesis

Through comparison with previous studies, our results provide insight into the latitudinal variation in crustal thickening along strike of the southern Central Andes. In the southern Puna plateau segment ($\sim 24.5\text{--}27.5^\circ\text{S}$), igneous La/Yb and Sr/Y trends recognized by Haschke et al. (2002) and quantified by recent paleomohometry (Bowman, 2024) show Mesozoic–Cenozoic crustal thickening, reaching $\sim 40\text{--}45$ km by ca. 100 Ma and $\sim 55\text{--}60$ km by ca. 40 Ma (Figure 9c). Farther south in the Aconcagua-Maipo region ($\sim 32.5\text{--}34.5^\circ\text{S}$), similar trends reported by earlier workers (e.g., Kay et al., 2005) and quantified by recent zircon and whole-rock paleomohometry (Carrapa et al., 2022; Muñoz-Gómez et al., 2025) indicate ≤ 5 km of Late Paleogene crustal thinning (ca. 40–25 Ma) followed by rapid thickening of the crust by ≥ 20 km to modern values of $\sim 50\text{--}55$ km between ca. 23 and 3 Ma (Figure 9c). Combined with our results, these data show that both the timing and magnitude of major crustal thickening decreased north-to-south along the southern Central Andes.

This trend aligns with spatiotemporal variations in fold-thrust belt development and crustal shortening (e.g., Giambiagi et al., 2022; Horton, 2018; Kley et al., 1999). At the latitude of the southern Puna plateau ($\sim 24.5\text{--}27.5^\circ\text{S}$), more than 250 km of shortening has accumulated progressively since the Late Cretaceous to early Cenozoic (Figure 9c; Carrapa et al., 2011; DeCelles et al., 2011; Giambiagi et al., 2022; Henriquez et al., 2022). By contrast, little to no Paleogene contraction is recognized south of $\sim 32.5^\circ\text{S}$, and reconstructed shortening magnitudes decrease rapidly southward to ≤ 46 km at $\sim 35^\circ\text{S}$ (e.g., Giambiagi et al., 2022; Howlett, 2025). This comparison suggests that a steep along-strike crustal thickness gradient emerged by Late Cretaceous–Paleogene time, when shortening was concentrated in northern segments.

Overall, our results show that variations in the Andean crustal root and the orogenic topography it supports closely track the distribution of tectonic shortening in space and time. They also reinforce models in which uplift and shortening are fundamentally coupled during Cordilleran orogenesis, rather than reflecting decoupled responses to lower crustal or lithospheric processes (e.g., Garzzone et al., 2017). The temporal correlation between geochemically resolved thickening and kinematically constrained retroarc shortening across multiple latitudes supports underthrusting of foreland lithosphere as a primary driver of Andean crustal growth, potentially augmented by associated processes such as enhanced arc productivity and magmatic addition.

7. Conclusions

Whole-rock paleomohometry and zircon petrochronology from a >70 Myr volcanic record in the Tinguiririca valley resolve the timing, magnitude, and drivers of crustal thickening during a key interval of Andean mountain building. Our principal findings are:

1. Trends in zircon Sm/Yb, Gd/Yb, and especially Dy/Yb ratios show general agreement with those of whole-rock mohometers, and both are consistent with geologic evidence for episodes of extension and shortening in the study area. We infer that these ratios preserve crustal thickness-sensitive melt source signals with greater fidelity than zircon Eu/Eu*, which records modification by decoupled magmatic processes. Although zircon (Sm, Gd, Dy)/Yb ratios can also be modified by shallow fractionation, these effects can be diagnosed by proxy divergence and correlation with Hf concentration. These findings support the use of these ratios to evaluate qualitative crustal thickness trends in ancient magmatic arcs.
2. Geochemical results indicate modest crustal thinning during the Late Cretaceous, coinciding with documented normal faulting and a transition to slow, oblique convergence. Eocene and Early Miocene samples yield similar crustal thicknesses (35.5 ± 3.9 and 35.6 ± 2.6 km, respectively), suggesting that extension documented elsewhere in the Abanico basin did not result in net crustal attenuation in the study area.
3. Crustal thickening of 10.4 ± 6.3 km (to 46.0 ± 5.7 km) occurred during the Neogene–Quaternary and was most rapid between ca. 17 and 13 Ma. Retroarc shortening likely served as the main trigger of this thickening by promoting underthrusting of foreland lithosphere. However, documented shortening magnitudes may be insufficient to account for the full thickening magnitude, suggesting contributions from processes such as magmatic addition or underplating of forearc material. In turn, shortening likely enhanced magmatic productivity by underthrusting melt-fertile material beneath the arc. Peak thickening was delayed relative to the Central Andes, mirroring trends in fold-thrust belt development.

These findings demonstrate that retroarc shortening can drive both rapid pulses of crustal thickening and magmatic flare-up in continental arcs. More broadly, the agreement among whole-rock paleomohometry, zircon

(Sm, Gd, Dy)/Yb ratios, and independent structural and geophysical constraints provides a transferable framework for reconstructing crustal thickness evolution in ancient orogens where such constraints are unavailable.

Inclusion in Global Research

This study was conducted by a team of researchers from the United States, Argentina, and Romania. C. Ronemus conducted portions of this research as a Fulbright Research Scholar at IANIGLA–CONICET in Mendoza, Argentina, where A. Echaurren and M. Barrionuevo are based; these collaborators contributed regional geologic expertise. We thank the Fulbright Commission, Miranda Grover, Laura Giambiagi, Julieta Suriano, José Mescua, Rodrigo Quiroga, Lucas Lothari, Ahmad Arnous, Macarena Bertoa, and others at CONICET who provided logistical support and scientific discussions during this exchange. This study builds on prior mapping and geochronology by Chilean and Argentine researchers, whose contributions are cited throughout the manuscript. We are committed to equitable research partnerships in the regions where we work.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

All geochemical data generated in this study are available in Supporting Information S1 (Data Sets S2 and S3) and are archived in the Zenodo repository at <https://doi.org/10.5281/zenodo.18423631> (Ronemus et al., 2026). No new software was created for this study. The Geochemical Arc Moho Estimator (GAME) software used for whole-rock paleomohometry was developed by Luffi and Ducea (2022b) and is archived in the OSF repository (Luffi & Ducea, 2022a).

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